

SDP 302

# Problem Set: Heteroscedasticity & Scale Transformation

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## Instructions & Information

The Engel data set is a historical benchmark data set in econometrics and statistics. Originating from Engel [1857](#) study on income and food expenditures of Belgian working-class families, it not only helped to establish *Engel's Law* but was also used in the foundational paper by Bassett and Koenker [1982](#) that shifted quantile regression methods from a purely theoretical estimation technique into a practical diagnostic framework. Because Engel's data shows that the variability of food expenditure grows as household income increases, this data set serves as a classic textbook example of heteroscedasticity, i.e. unequal variance in the error term.

In this exercise we will use Engel's data set to study the effects of heteroscedasticity on OLS regression. Based on standard economic theory we aim to fit the following simple model:

$$\text{foodexp}_t = \beta_0 + \beta_1 \text{income}_t + \epsilon_t$$

However, because heteroscedasticity violates the classical Gauss-Markov assumption of constant variance in the error terms, estimating the model above using OLS produces estimates that are unbiased but inefficient and less precise, alongside invalid standard errors. One way, to address these issues is to perform a logarithmic transformation which compresses the scale on which variables are measured. In our case, using double-log model, i.e. a log-log specification, yields:

$$\ln(\text{foodexp}_t) = \beta_0 + \beta_1 \ln(\text{income}_t) + \epsilon_t$$

Beyond addressing heteroscedasticity, using a log scale offers additional benefits by converting absolute changes into relative percentage terms. In the simple linear model above, the estimated  $\beta_1$  coefficient denotes the unit change in food expenditure for every one-unit change in income. However, in the double-log model, coefficient  $\beta_1$  represents the income elasticity of demand, i.e. a 1% increase in household income yields an estimated  $\beta_1\%$  change in food expenditure. This parameter can be directly used to test *Engel's Law*, which posits that because food is a necessary good, its income elasticity of demand lies between 0 and 1.

While log transformations can solve the issue of heteroscedasticity, they introduce several major practical and statistical drawbacks. First, logarithms are only defined for positive values. If the data set contains zeros or negative values, these values will automatically be dropped from estimation. Second, any values very close to zero will drop toward negative infinity when log-transformed. This turns tiny non-zero observations into influential outliers, pulling the OLS regression line and distorting parameter estimates. Third, while taking logarithms reduces variance for large values of the independent variable, it can over-compress lower values and introduce artificial heteroscedasticity at lower levels. Fourth, according to Jensen [1906](#), the expected value of a logarithm is not equal to the logarithm of an expected value. Hence, taking logs will distort means when expected log values will be transformed back to their original scale.

## References

- Bassett, Gilbert and J. Roger Koenker (1982). “Robust Tests for Heteroscedasticity Based on Regression Quantiles”. In: *Econometrica* 50.1, pp. 43–62.
- Engel, Ernst (1857). *Zeitschrift des Statistischen Bureaus des Königlich Sächsischen Ministeriums des Innern*.
- Jensen, J. L. W. V. (1906). “Sur les fonctions convexes et les inégalités entre les valeurs moyennes”. In: *Acta Mathematica* 30, pp. 175–193.

### [EXR] Exercise 1 (OLS & Heteroscedasticity: The Engel Data Set)

The Engle data set is part of Python’s `statsmodels` library and can easily be loaded calling the method `sm.datasets.engel.load_pandas().data`. Import the data into a `DataFrame` and answer the following questions:

- Fit the simple linear OLS model. Do you notice any surprising results such as unexpected p-values or coefficient signs? What might explain these findings? Provide a clear, meaningful interpretation of the estimated parameters in the context of household budgets.
- Visualise the data set using a scatter plot, overlaying the estimated regression line. Do you observe any patterns or systematic deviations from the classical Gauss-Markov assumptions?
- Conduct both a Breusch-Pagan and a White test for heteroscedasticity. Provide a clear interpretation of the test statistics and p-values.
- Re-run the Breusch-Pagan and White tests using the residuals from the log-transformed model. How do these results compare with your findings from part c), and what does this imply about the effectiveness of the log transformation??

### Solution:

[Dashboard](#)